■ NEUROSCAN AI

DIAGNOSTIC REPORT

Brain MRI Analysis — AI Assisted

Report ID:	RPT-A693ADD2	**Generated:**	2026-04-13 11:26:03 IST
Patient:	jo	**Study Date:**	April 13, 2026
Patient ID:	123	**Modality:**	MRI Brain (AI Inferred)
Model:	BrainTumorAI v2.0 Ensemble	**Model AUC:**	0.9996 | Acc 98.78%

PRIMARY FINDING

GLIOMA

Diagnosis: GLIOMA
Confidence: ■■■■■■■■■■■■■■■■■■■■ 81.1%
Risk Level: ■ CRITICAL
Processing: 16.94s (TTA 6-view ensemble)
Method: EfficientNet-B3 + ResNet50+CBAM + DenseNet121

DIFFERENTIAL DIAGNOSIS

Class	Probability	Bar Chart	Notes
Glioma	**81.1%**	■■■■■■■■■■■■■■■■	*← Primary*
Meningioma	**6.6%**	■■■■■■■■■■■■■■■■	
No Tumor	**6.6%**	■■■■■■■■■■■■■■■■	
Pituitary	**5.7%**	■■■■■■■■■■■■■■■■	

ENSEMBLE AGREEMENT

Model	Confidence	Bar
EfficientNet-B3	**96.1%**	■■■■■■■■■■■■■■
ResNet50+CBAM	**47.5%**	■■■■■■■■■■■■■■
DenseNet121	**94.7%**	■■■■■■■■■■■■■■
Ensemble (TTA)	**81.1%**	**← FINAL**

IMAGING EVIDENCE

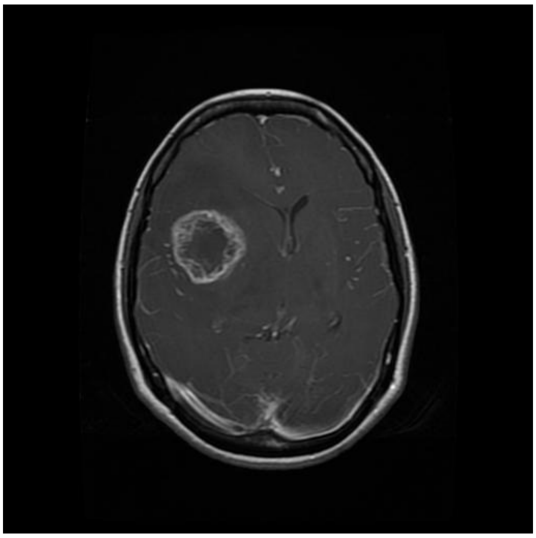

Figure 1: Original MRI scan

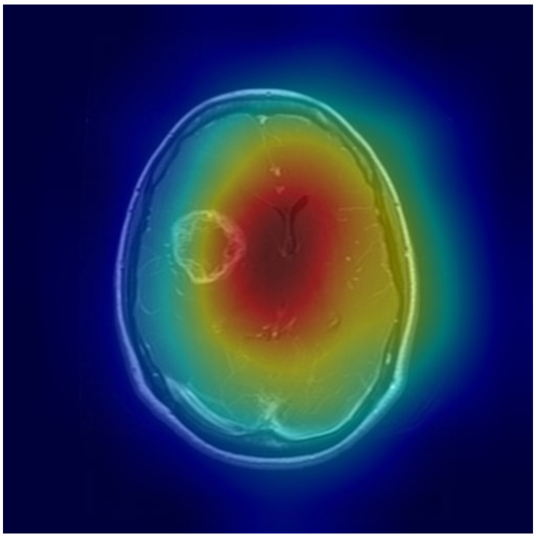

Figure 2: Grad-CAM activation overlay

GRAD-CAM ANALYSIS PANEL

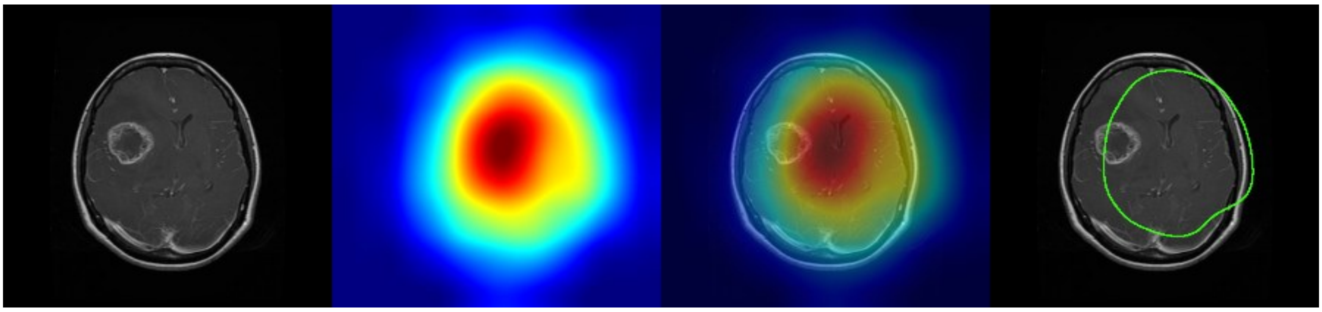

Figure 3: (Left→Right) Original MRI | Activation Map (Jet) | GradCAM Overlay | Tumour Boundary (threshold > 0.5)

■ Red areas = high tumour probability ■ Yellow = moderate ■ Blue = low activation

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TUMOUR LOCALISATION (GRAD-CAM ANALYSIS)

Parameter	Value	Notes
Estimated Area	19.3% of visible scan	Moderate
Hemisphere / Location	CENTRAL	Based on centroid x-position
Centroid Position	(289, 251) pixels	Hottest activation pixel centroid
Bounding Region	[177,111] → [409,391]	Pixel coordinates of activation box
Activation Score	70 / 100	Mean activation in region (0=low, 100=high)
GradCAM Computed	5.18s	CPU inference time

CLINICAL INTERPRETATION

The AI model identifies imaging features consistent with a glial neoplasm. The Grad-CAM localisation map highlights a region of interest with high activation intensity. Gliomas represent the most common primary brain tumours and vary widely in grade (WHO I–IV). Immediate referral to neurosurgery and neuro-oncology is strongly recommended. MRI with contrast enhancement, spectroscopy, and perfusion imaging should be obtained. Biopsy and molecular profiling (IDH status, MGMT methylation) may be required for definitive grading.

RECOMMENDATIONS

1. Urgent MRI brain with gadolinium contrast enhancement
2. Neurosurgical consultation within 48 hours
3. Neuro-oncology multidisciplinary team review
4. Biopsy / resection planning (IDH, MGMT, EGFR profiling)
5. Baseline neuropsychological assessment

TECHNICAL DETAILS

Parameter	Value
Architecture	EfficientNet-B3 + ResNet50+CBAM + DenseNet121
Inference	TTA (6-view: orig, H-flip, V-flip, rot90/180/270)
Ensemble method	Confidence-weighted average + temperature scaling
GradCAM method	Standard Grad-CAM on last conv layer per backbone
Model AUC	0.9996 (macro one-vs-rest, test set n=1311)
Model Accuracy	98.78%
Sensitivity	98.68% | Specificity: 99.59%
Device	CPU (no GPU detected)
Processing time	16.94s (inference) + 5.18s (GradCAM)